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March 18, 1965

SUPERSEDING
Int. Fed. Spec. UU-B-0038a (GSA-FSS)
August 10, 1964

FEDERAL SPECIFICATION

BAGS, PAPER, (HARDWARE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for hardware paper bags of standard sizes and weights of paper stock that are in general demand. The minimum basis weights and strength of paper are also established.

1.2 Classification.

1.2.1 *Sizes.* The size (capacity) of the bag shall conform with the standard bag numbers listed in table I, or as specified (3.3.2 and 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-31—Paper; General Specifications and Methods of Testing.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-P-271—Paper, Wrapping, Waterproofed, Kraft.
UU-S-48—Sacks, Shipping Paper.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The paper used in the fabrication of the hardware bags shall consist of 100 percent unbleached sulphate (kraft) fiber and shall conform to the applicable requirements in table I. The paper shall be resin-sized.

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3.1.1 *Bags of other weights of paper.* When this specification is cited in procurements for hardware bags with weights of paper other than those specified in table I, the paper shall have a minimum average bursting strength in points and a minimum average tearing resistance, machine direction, in grams which are the same percentages of the basis weights specified, as are the corresponding strength requirements for the next lighter basis weight paper listed in table I or II (see 6.1).

3.2 **Construction.** Hardware bags shall be so constructed that all seams shall adhere securely throughout their area.

TABLE I. *Self-opening (automatic) bags, standard bag numbers, capacities, and weights of paper*

Bag number	Capacity[1] (tolerance minus 2 percent)	Basis weight[2] 24 by 36 —500 (tolerance —5 percent)	Average bursting strength minimum	Average tearing resistance, machine direction (minimum)
	Cubic inches	*Pounds*	*Points*	*Grams*
3	103	50	31	92
5	137	50	31	92
10	255	60	38	121
16	490	60	38	121
20	542	60	38	121
25	584	60	38	121

[1] The capacity shall be determined by the method of measuring bags and computing their cubic capacity as specified in 4.3.1.2.1.

[2] See 4.4.1.

3.3 **Dimensions.**

3.3.1 The dimensions of the standard bag numbers as produced by different manufacturers may vary somewhat and shall be considered acceptable provided that such bags meet the applicable minimum cubic inch capacity as specified in table I.

3.3.1.1 The approximate dimensions of hardware bags are furnished in table II as a guide only. The dimensions of the standard bag numbers as produced by different manufacturers may vary somewhat and shall be considered acceptable provided that such bags meet the applicable minimum cubic inch capacity as specified in table I.

TABLE II. *Approximate dimensions (see 3.3.1)*

Bag number	Face width	Tuck (bellows)	Length (height)
3	4-3/4	2-3/4	7-7/8
5	5	3-1/8	8-3/4
10	6-1/4	3-7/8	10-1/2
16	8-3/16	5-3/8	11-1/8
20	8-3/16	5-3/8	12-5/16
25	8-3/16	5-3/8	13-1/4

3.3.2 *Bags (of sizes other than specified).* When this specification is cited in procurement of hardware bags of sizes other than those specified in tables I and II, a tolerance of minus 2 percent will be permitted in the capacity specified (6.1).

3.4 **Bag marking.** Unless otherwise specified, bags shall be imprinted with the following identification markings, as applicable.

3.4.1 *Bag number.* The appropriate bag number as shown in column 1 of table I shall be placed on each bag, as applicable, as part of the manufacturer's trademark or imprint.

3.4.2 *Size (capacity).* The letters SS, or the industry's emblem SS, shall be marked or imprinted on each bag when the bag conforms to the applicable standard capacities specified in table I. If marking of each bag is impracticable, only each package as specified in table IV is required to be marked as specified herein.

3.4.3 *Size (capacity) and weight.* The letters SWS, or industry's emblem SWS, shall be marked or imprinted on each bag when the bag conforms to the applicable standard capacity and corresponding standard weight of paper as specified in table I. If marking of each bag is impracticable, each package, as specified in table IV, is required to be marked as specified herein.

3.5 **Workmanship.** Hardware bags shall be uniformly constructed and shall be free

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from holes, tears, cuts, splits, or other defects which might impair their usefulness, and shall conform to the levels of quality established herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling for inspection and tests.* Sampling for inspection and tests shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all material of the same size.

4.2 Inspection of materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects set forth in the applicable examination paragraphs at the inspection levels and acceptable quality levels (AQL's) in 4.3.1.5. No more than one bag, randomly selected, shall be drawn from any one package from each lot for each weight and size offered for acceptance for visual and dimensional characteristics. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of packages (see table IV) of the applicable quality for each weight and size bag, for examinations under 4.3.1.1, 4.3.1.2, 4.3.1.3, and in units of bales for examination under 4.3.1.4.

4.3.1.1 *Examination of the end item for defects in construction and workmanship.* The sample unit for this examination shall be one bag.

Examine	Defect
Appearance	Surfaces not clean; presence of any foreign matter or dirt. Tear, cut, puncture, split, slit (other than those necessary to make bag i.e., slitter cuts), or hole; chafed spot or scuff mark which definitely weakens the paper.
Construction	Seams poorly glued, do not adhere securely throughout their entire area. Bag improperly folded to extent end use is impaired. Construction of bag not in conformance with contract requirements for the bag (see figure 1).
Identification marking (as applicable)	Omitted, illegible, or not the applicable bag number or emblem.

4.3.1.2 *Examination of the end item for dimensional defects.* The sample unit for this examination shall be one bag. (Methods of measuring the bag and computing their respective capacities shall be performed in accordance with 4.3.1.2.1.)

Examine	Defect
Special sizes (when so specified)	Any dimension less than specified by more than 1/8 inch from size specified in contract or order.
Capacity	More than 2 percent below the contractual requirements for the specified bag number (see tables I and II).

4.3.1.2.1 *Methods of measuring the bags and computing their capacity.* A scale graduated in eights of an inch shall be used in marking the measurements.

4.3.1.2.1.1 *Self-opening (automatic) bags (fig. 1).* Position of bag—Bottom to the

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front or rear. Point of measurement—Dimension lines in diagrams.

Point of measurement—Dimension lines in diagrams.

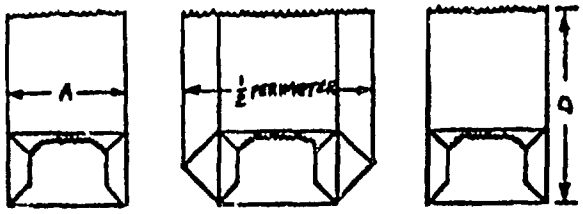


FIGURE 1. Method of measuring self-opening (automatic) bags.

4.3.1.2.1.1.1 *Face width, A.* Measure across the front of the folded bag just above the top of the bottom fold.

4.3.1.2.1.1.2 *Bottom width, C.* Spread the bag so that the tucks normally behind the face are fully extended outward. Press the bag flat and measure full width as close to the bottom fold as possible. Subtracting the face width from this measurement gives dimension C.

4.3.1.2.1.1.3 *Bag length, D.* Measure along center of the front face of the bag, from the lower bottom fold to the top of the serrated edge.

4.3.1.2.1.1.4 *Cubic capacity.* Cubic capacity (A x C x D) is the product of the face width, bottom width, and bag length.

4.3.1.3 *Examination of end item for count per package.* The sample unit for this examination shall be one package of bags. The average count per package shall be not less than the specified quantity (see table IV).

4.3.1.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and markings comply with the requirements of section 5. The sample unit shall be one bale, fully packed, which may be selected just prior to the closing operations. Bales fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Not in accordance with contract requirements; packages not wrapped and adequately secured as specified. Mixed sizes in unit package.
Packing	Not level specified; not in accordance with contract requirements. Arrangement and number of packages per bale not as specified (see table IV). Wrapping papers (as applicable) omitted, damaged, or not type and grade specified. Bale material not as specified; closures not accompanied by specified or required methods or materials.
Markings	Illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.

4.3.1.5 *Inspection levels and AQL's for examinations.* The inspection levels, for determining the sample size, and the AQL's expressed as defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL's
4.3.1.1	S-3	4.0
4.3.1.2	S-2	2.5
4.3.1.3	S-2	—
4.3.1.4	S-2	4.0

4.3.2 *Testing of the end item.* The end item shall be tested for the characteristics indicated in table III for each lot presented for examination for each type, grade, size, and weight of material. The sample unit specimens, the amount required to perform all tests for one sample unit, shall be as indicated in 4.3.2.1. No more than one sample unit shall be drawn from any one bale. Five sample units, randomly selected throughout the lot, shall be tested with no evidence of failure to meet the specified requirements. Each sample unit shall be represented equally for tests using composite samples.

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TABLE III. *Instructions for testing*

Characteristic	Specification reference	Requirements applicable to		Number determinations per sample unit[2]	Results reported as	
	[1]	Individ. unit	Lot aver.		Pass or fail[3]	Numerically to nearest[4]
Stock	3.1	—	—	Average of 2 (composite)	—	5 percent
Sizing	3.1	X	—	2 (composite)	X	—
Basis weight: (24 by 36—500)	3.1	—	X	6 sq. ft. minimum area	—	0.1 pound
Other weights	3.1.1	—	X		—	0.1 pound
Bursting strength:	3.1	—	X	Average of 10 (5 each side)	—	point
Other weights (as specified)	3.1.1	—	X		—	point
Tearing resistance. (machine direction)[5]	3.1	—	X	Average of 5	—	gram
Tearing resistance (machine direction)[5] (other weights (as specified)	3.1.1	—	X	Average of 5	—	gram

[1] Test methods, as referenced in 4.4.1 shall be performed in accordance with UU-P-31, except as the test methods described in UU-P-31 may conflict with this specification. In case of conflict, this specification shall govern.

[2] The composite sample shall consist of representative portions taken from each sample unit and combined into a single sample.

[3] If failure is indicated, report either description or numerical point of failure, as applicable.

[4] Test reports include all values on which results are based.

[5] Tearing resistance measurements shall be made in accordance with method 170 of UU-P-31, utilizing the Elmendorf tearing strength tester having sector of the original or unmodified shape. Disregard that portion of method 170 which may direct eliminating tearing strength readings when the specimen rubs against this sector.

4.3.2.1 *Sampling for tests.* The number of specimens (bags) applicable to the sample unit for each of the different weight and size of the end product shall be as follows:

Bag number (size)	Sample unit (no. of bags per sample unit)
3	18
5	8
10	5
16, 20, 25	3

4.4 **Tests.**

4.4.1 *Test methods.* The following tests shall be conducted in accordance with methods described in UU-P-31:

Equipment	Test method
Atmospheric conditions	102
Basis weight	110
Bursting strength	112
Tearing resistance	170
Rosin sizing (qualitative)	250
Fiber identification and quantitative determination	300

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A.

5.1.1 *Level A.* Bags of one size only, in

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the quantity specified in table IV, shall be stacked flat, in accordance with the manufacturers' commercial practice, and assembled into package. The package shall be enclosed in a sleeve or wrap fabricated from kraft paper having a basis weight of 60 pounds per ream (24 x 36—500) conforming to grade B of UU-P-268. String tying of the bags within the package is optional. If string tying is used, the quantity per tie should be as specified in table IV.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Bags of one type, grade, and size only, packaged as specified in 5.1.1, shall be packed in quantity specified in table IV and shall be compactly arranged and baled by wrapping with two sheets of kraft wrapping paper, each sheet having a basis weight of 60 pounds per ream (24 x 36—500) conforming to grade B of UU-P-268. In addition, an overwrap of waterproofed kraft wrapping paper conforming to grade B-2 of UU-P-271 shall be applied. Alternatively, the paper used for wrappings to form a bale shall consist of a single inner wrap of kraft wrapping paper having a basis weight of 60 pounds per ream (24 x 36—500) conforming to grade B of UU-P-268 and an intermediate and outer wrap of wet strength heavy duty kraft paper having a basis weight of 60 pounds per ream (24 x 36—500) conforming to grade B of UU-S-48. All paper wrappings shall overlap, widthwise and lengthwise, a minimum of 4 inches. Each completed bale shall then be securely cross tied with twine or cotton rope, with one cross tie over the center of each stack, with the twine or cotton rope knotted at each crossover to prevent loosening or untying. The twine or cotton rope shall have a breaking strength of not less than one and one-half times the weight of the finished bundle (see 6.4).

5.2.2 *Level B.* Bags of one size only, packaged as specified in 5.1.1, shall be packed in bales as specified in 5.2.1 except that only two sheets of kraft wrapping paper will be required on a bale. The kraft wrapping paper shall have a basis weight of 60 pounds per ream (24 x 36—500) conforming to grade B of UU-P-268. The completed bale shall be cross tied as specified in 5.2.1.

5.2.3 *Level C.* Bags of one size only, packaged as specified in 5.1.1, shall be packed in the quantity specified in table IV, in shipping containers that comply with the applicable carrier rules and regulations for safe transportation to point of destination specified in shipping instructions.

5.3 **Marking.**

5.3.1 *Civil agencies.* Bales shall be marked in conformance with Fed. Std. No. 123.

5.3.2 *Military agencies.* Bales shall be marked in accordance with MIL-STD-129.

TABLE IV. *Packaging and packing quantities*

Bag number	Packaging			Packing	
	Quantity of bags (see 5.1.1)			Quantity per bale	
	Reverse every	Banded every	Per package	Bags	Packages
3	50	500	500	2,000	4
3	50	500	500	3,000	6
5	50	500	500	2,000	4
5	50	500	500	3,000	6
10	50	500	500	1,000	2
16	50	500	500	500	1
20	50	500	500	500	1
25	50	500	500	500	1

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Bag number (size) of bag required (1.2.1 and tables I and II).

(*c*) When bags of other weights of paper are required (3.1.1).

(*d*) When bags of other sizes are required (3.3.2).

(*e*) Inspection responsibility methods, and tests (section 4).

(*f*) Applicable levels of packing (5.2).

6.2 Intended use. Hardware bags are heavy weight to provide extra strength to accommodate irregular and heavy hardware items.

6.3 Exceptions to specification. Nothing in this specification shall be construed to prohibit the use of bags of special design, sizes, or heavier weights of paper when, in the experience and judgment of the purchaser, the nature of the items intended to be packed, or the usage involved, justifies such bags.

6.4 Certificate of compliance. At the option of the contracting officer the twine or cotton rope may be accepted on the basis of a contractor's certificate of compliance for the breaking strength requirements specified in 5.2.1 of this specification.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bags, paper, not otherwise indexed by name.
Carload minimum weight, 36,000 pounds.

Motor:

Bags, paper, not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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